SAMPLE IDENTIFICATION	
Code	Faro 32
Well	2-LFST-1-AM
Depth	2.156,72 m
Coordinates (poço)	Lat. 1°57'20,02" S | Long. 56°58'17,00" W
Shapetion / Basin	Faro / Basin do Amazonas
Age	Tournaisiana–Viseana (Carbonífero Inferior)

GENERAL DESCRIPTION OF THE THIN SECTION
Coarse grain size, with well-individualized grains and distinct outlines. Most striking feature: abundance of brownish to orange intergranular material (more pronounced ferruginous cementation). There are no fractures or grading. Isolated grain with vivid interference colors (SW sector) consistent with mica. Primary porosity eliminated.

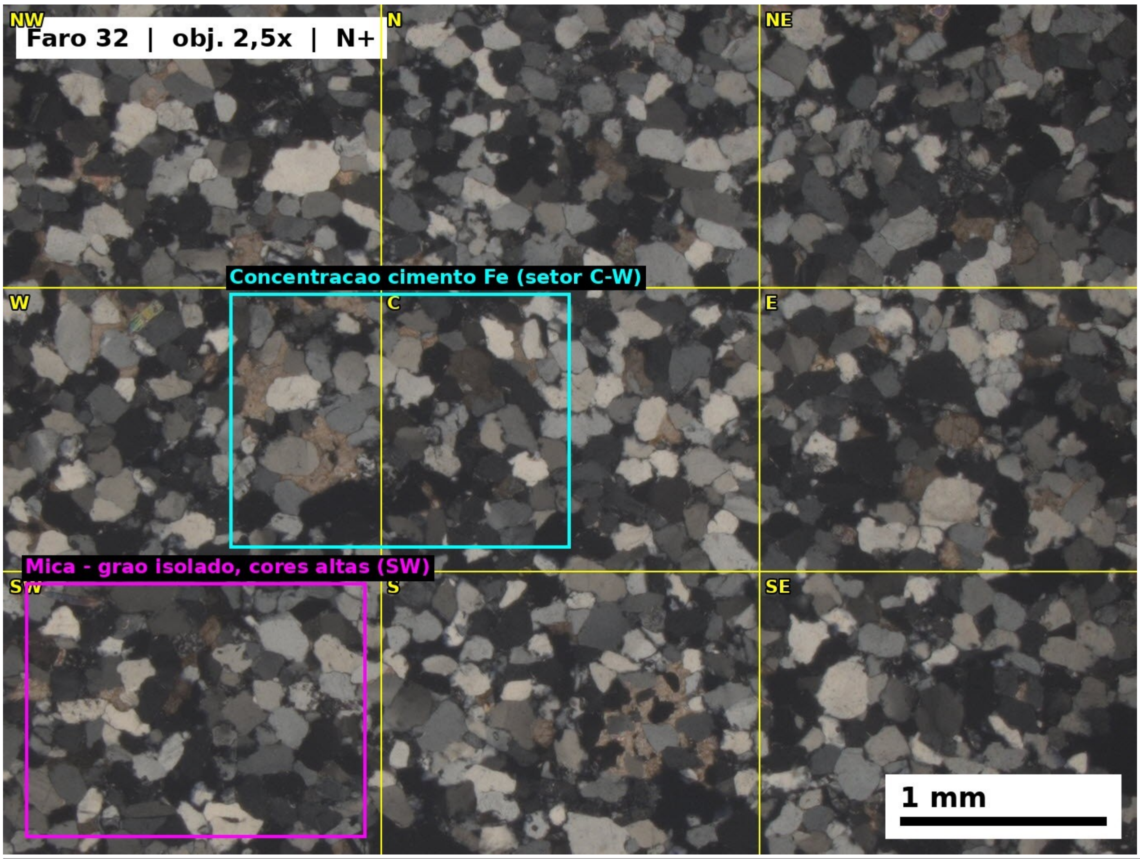
ESTIMATED MODAL COMPOSITION (% vol)	
Quartz monocrystalline	~65–70%
Quartz polycrystalline	~5–8%
Feldspar (K-feldspato + plagioclase)	~5–8%
Fragments lithic	~3–5%
Micas (muscovite ± biotite)	~1–2%
Minerals opaque	~2–3%
Cement (ferruginous + siliceous)	~8–12%
Matrix clayey	~2–3%
Porosity visible	<1%

Composition: Dominant quartzose composition: quartz arenite to sublitharenite. Coarse grain size. Grain with high birefringence may correspond to muscovite (mica).

MICROSCOPIC DESCRIPTION	
Textura	Clastic terrigenous, grain-supported, grains well-individualized, texture homogeneous.
Grain size	sand medium a coarse (~0,25–0,70 mm). Mode: areia medium (~0,35–0,50 mm).
Grain-size sorting	Moderately sorted.
Packing and contacts	Dense. Long (~45%), concave-convex (~35%), point (~15%), sutured (~5%). Lower % sutured despite greater depth (~2,157 m) (coarse grain size reduces contact area); early ferruginous cementation may have inhibited pressure/dissolution.

Cement / diagenesis	Abundant to very abundant ferruginous cementation (hematite/goethite) filling interstices and forming grain-coatings. Subordinate siliceous cementation (Qz overgrowth) associated with sutured contacts. Fe abundance may reflect: (1) greater original permeability; (2) greater availability of Fe due to dissolution of mafic minerals.
Porosity	No porosidade primária aberta evident. Interstícios filled por cimento ferruginous.
Maturity	Textural: mature. Compositional: mature (>65% Qz). Compatible with shoreface/foreshore reworked by storm waves.

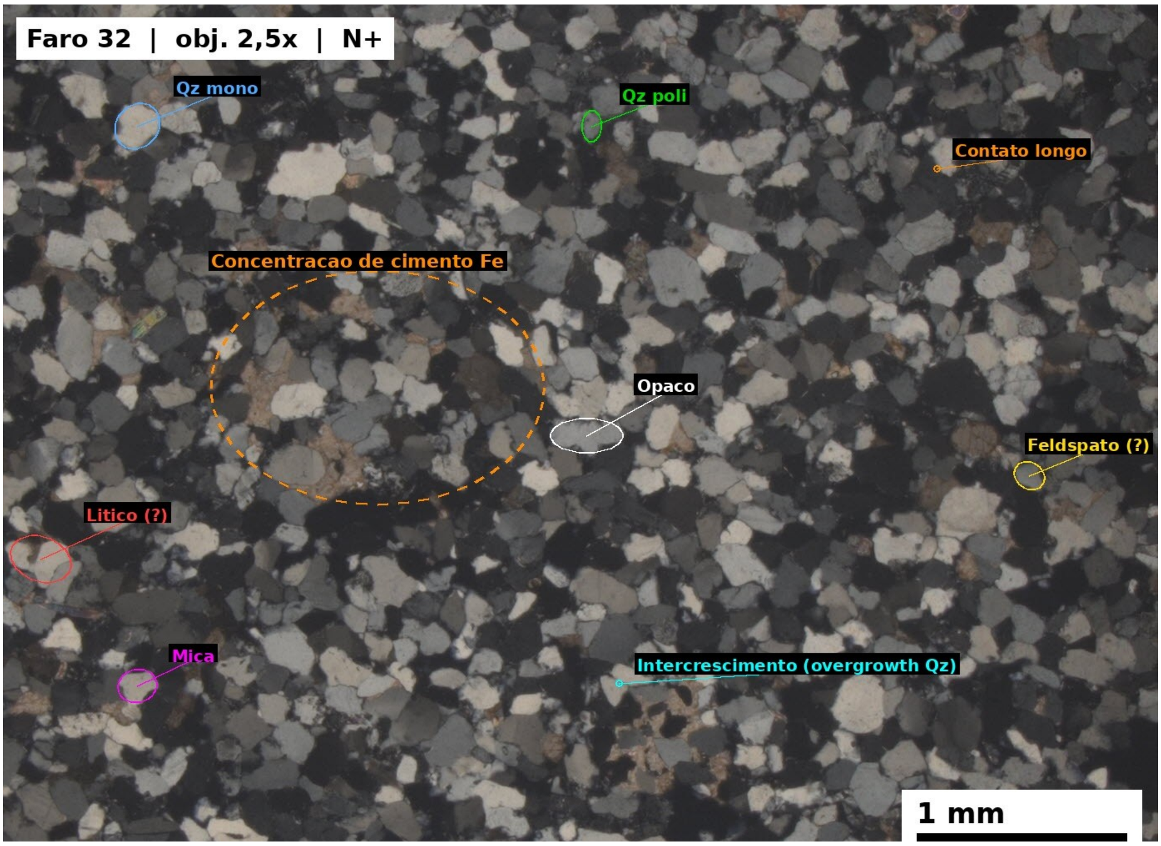
GRAIN SECTOR ANALYSIS (3×3 grid)					
Sector	**Mineral**	**Size (mm)**	**Sphericity**	**Roundness**	**Shape**
NW	mono Qz	0,45	High	Subrounded	Equidimensional
NW	mono Qz	0,30	Medium	Rounded	Equidimensional
N	mono Qz	0,50	High	Rounded	Equidimensional
N	Feldspar (?)	0,35	Medium	Subangular	Prismatic
N	poly Qz	0,40	Low	Subangular	Elongated
NE	mono Qz	0,35	High	Subrounded	Equidimensional
W	mono Qz	0,40	High	Rounded	Equidimensional
W	Lithic	0,50	Low	Subangular	Elongated
C	mono Qz	0,55	High	Rounded	Equidimensional
C	Opaque	0,15	High	Rounded	Equidimensional
E	mono Qz	0,45	High	Subrounded	Equidimensional
E	Feldspar	0,30	Medium	Subangular	Prismatic
SW	mono Qz	0,35	Medium	Subrounded	Equidimensional
SW	Mica	0,25	Low	—	Discoidal
S	mono Qz	0,40	High	Rounded	Equidimensional
S	mono Qz	0,25	Medium	Subrounded	Equidimensional
SE	mono Qz	0,50	High	Rounded	Equidimensional
SE	poly Qz	0,35	Low	Subangular	Elongated



Grain population: Dominant population: mono Qz, high sphericity, subrounded/rounded, equidimensional, 0.35–0.50 mm. Larger and more rounded grains than in well 1-GU-2-PA → greater transport/reworking. Subordinates: poly Qz, rare lithics/feldspars, isolated mica.

CLASSIFICATION
Medium to coarse quartz arenite to sublitharenite, moderately sorted, mature, abundant Fe cement.

PHOTOMICROGRAPH



Photomicrograph (N+, 2.5× objective) of medium to coarse quartz arenite to sublitharenite, grain-supported. Dominant monocrystalline quartz (mono Qz); subordinate polycrystalline quartz (poly Qz), feldspar (F), and lithic fragments (L); isolated mica grain identified by high-order interference colors. Abundant ferruginous cementation (the most pronounced of all thin sections), associated with predominantly long contacts.